

# Reproduction of atong01/conditional-flow-matching: OT-CFM CIFAR-10 FID

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*Submitted by K-Veritas Team*

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Independent reproduction · Artifact evaluation · Computer vision

## Abstract

Evaluated the authors' released OT-CFM CIFAR-10 weights (400,000 training steps, release 1.0.4) with the repository's compute\_fid.py and the README settings at the pinned commit: 50,000 samples generated with the adaptive dopri5 solver (tolerance 1e-5) and FID computed against the CIFAR-10 training set with clean-fid. Training was not repeated (400,000 steps exceeds the time budget); the weights file is attested by hash in the report. SciPy was pinned below 1.16 for clean-fid compatibility. One NVIDIA RTX 5060 Ti.

### About this record

Title, authors and abstract were provided by the authors and reviewed for completeness, not for scientific soundness. Every number in this record is copied from the signed data of the sealed reports [2], which remain authoritative.

This cover page is not signed. Check it against the SHA-256 listed at <https://kveritas.org/records/2610.00018v1>, and verify each report with its code bundle: `kveritas verify <report.pdf> --bundle <bundle.zip>`.

## 1 Sealed reports

The experiments were sealed with K-Veritas [1], which binds each result to the code, hardware and time that produced it.

| Report | Label                     | Sealed     | Runs | Invocations | Signer                         |
|--------|---------------------------|------------|------|-------------|--------------------------------|
| r1 [2] | OT-CFM CIFAR-10 FID (50k) | 2026-10-07 | 1    | 1           | Signed by the K-Veritas server |

## 2 Results

### r1: OT-CFM CIFAR-10 FID (50k)

| Run | Metric  | Final value | Step |
|-----|---------|-------------|------|
| 1   | fid_50k | 3.64607     |      |

| Run | Inline claim | Value   | Phase |
|-----|--------------|---------|-------|
| 1   | fid_50k      | 3.64607 |       |

### 3 Setup

| Report | Item             | Value                                           |
|--------|------------------|-------------------------------------------------|
| r1     | Operating system | linux / amd64                                   |
| r1     | CPU              | Intel(R) Xeon(R) W-2145 CPU @ 3.70GHz, 16 cores |
| r1     | GPU              | NVIDIA GeForce RTX 5060 Ti, 16311 MiB           |
| r1     | Memory           | 31.0 GB                                         |
| r1     | Total run time   | 7437.0 s                                        |

### 4 Files

Each report and code bundle must hash to these values. A bundle belongs to the report whose signed data names its hash.

| File            | SHA-256                                                          | Size    |
|-----------------|------------------------------------------------------------------|---------|
| r1.pdf          | 42a4d34dc81b7cd72051c5a169cb6a432e1067860a7ee276d2d33edb9650e662 | 6.8 MB  |
| r1.kvbundle.zip | 8d2399cb0fe250bae6ec09526af5f450e2875a02ce648669f0657b371dbe60f1 | 19.0 MB |

### 5 Links and licence

Paper: <https://arxiv.org/abs/2302.00482>

Reproduction of:

<https://github.com/atong01/conditional-flow-matching/tree/7c653857de4c25b979a740ef8b6a5d5ddb09b6c1>

Record: CC-BY 4.0. Code: MIT, as declared by the authors. Tags: flow-matching, generative-models, cifar-10, fid.

Listed, not endorsed. Review checks the artifact, never the science. Reports are never edited; changing them makes a new version. Other fields are corrected in place, and every correction is logged in metadata.yaml.

### References

[1] M. K. Keita and C. Homan. Computer Science Conferences Should Require Nonrepudiable Experimental Results. NeurIPS 2026 Position Paper Track. arXiv:2605.08586.

[2] Sealed experiment report r1 (OT-CFM CIFAR-10 FID (50k)), kv:2610.00018v1/r1. Data hash 45245b62b22d14bef244aff320dcb53ce6466f333abdaa5ae41a55bd7e184562.  
<https://kveritas.org/records/2610.00018v1/r1.pdf>